

Student Takeaway for Sync Session

<<Week 3: Calculus Examples and Functions in Machine Learning>>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Calculus Examples and Functions in Machine Learning	
Session duration	1.5 hours	
Session type	Lectures: Understanding differentiation, gradients, optimisation, and their role in ML. Case Studies: Practical applications of calculus in machine learning, including gradient descent, loss functions, and activation functions.	
Scope	This session introduces key calculus concepts and their applications in ML: - The role of differentiation and integration in ML. - How derivatives and gradients optimise ML models. - The importance of backpropagation, regularisation, and feature transformation.	
Learning objectives	Objective	Core capability
	Understand differentiation and integration in ML	Analytical thinking in optimisation
	Apply gradient descent for model optimisation	Practical experience in parameter tuning
	Implement backpropagation using chain rule	Understanding of neural network training
	Explore the role of activation and loss functions	Problem-solving using calculus in ML
	Connect theoretical concepts to real-world ML applications	Hands-on application in ML algorithms
Software/Tools	- Python (Libraries: NumPy, TensorFlow, PyTorch, Scikit-learn, Matplotlib) - IDE: Jupyter Notebook - Dataset: Various ML examples (e.g., optimisation, probability distributions) - Presentation Tool: PowerPoint	

Pro tips for success:

- Ask Bold Questions:** No question is too small curiosity drives learning!
- Be Hands-On:** Experiment with code, tweak parameters, and observe changes.
- Collaborate:** Share ideas in discussions—your perspective can spark insights!

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1	Introduction	Overview of session objectives.	Why does ML require calculus?
2, 3	Understanding Calculus	Learn about differentiation and integration.	Example: Cost function minimisation
4, 5	Derivatives and Gradients	Understand how functions change and their ML applications.	Example:
6-8	Partial Derivatives	Learn how partial derivatives work in ML models.	Example: Gradient computation in multi-variable functions
9-10	Hessian Matrix & Optimisation	Understand second-order derivatives and their role in optimisation.	Example: Newton's method
11-13	Integration in Probability	Learn how integration applies to probability distributions.	Example: Area under a probability density function
14-15	Jacobian & Hessian in Neural Networks	Explore derivatives in neural network training.	Example: Optimising deep learning models
16-17	Gradient Descent	Discover how models learn iteratively.	Example: Parameter tuning in logistic regression
18-20	Backpropagation & Chain Rule	Understand weight updates using calculus.	Example: Chain rule in deep learning
21-23	Regularisation & Loss Functions	Learn how differentiation prevents overfitting.	Example: L1 and L2 regularisation
24-26	Convex vs. Non-Convex Functions	Explore function properties in optimisation.	Example: MSE vs. deep learning losses
27-29	Summary & Closing	Recap key concepts and applications.	Reflect on real-world ML applications using calculus.

Post session

Reflection challenge	What did you find exciting about decision trees?
Explore more	• Read: Scikit-learn documentation on Decision Trees and hyperparameter tuning. • Watch: YouTube videos on decision tree applications in fields like marketing and healthcare. • Play around: Experiment with larger datasets (try Kaggle!) to see how decision trees scale.
Get inspired	Did you know that decision trees help banks detect fraud and doctors diagnose diseases? Imagine the possibilities when you master this skill!

The journey ahead

Next, dive into Essential Concepts in Probability and Statistics, where you'll learn about probability distributions, statistical inference, and their role in ML algorithms.

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